BY ORDER OF THE COMMANDER
AIR COMBAT COMMAND

AIR COMBAT COMMAND MANUAL
11-2E-9, Volume 2

13 NOVEMBER 2023

Flying Operations

E-9—AIRCREW EVALUATION CRITERIA

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

ACCESSIBILITY: This publication is available for downloading or ordering on the e-Publishing website at **www.e-Publishing.af.mil**.

RELEASABILITY: There are no releasability restrictions on this publication.

OPR: ACC/A3TV

Certified by: ACC/A3T
(Mr. John Costa)

Supersedes: ACCMAN 11-2E-9V2, 1 October 2019

Pages: 23

This volume implements AFMAN 11-202, Volume 2, *Aircrew Standardization and Evaluation Program*. It provides guidance and procedures on conducting an aircrew evaluation, applicable to all units assigned the E-9 aircraft, to accomplish the mission safely and successfully. This publication applies to all E-9 aircrew under ACC authority, including Air Force Reserve or Air National Guard personnel that operate E-9 aircraft. This publication does not apply to the United States Space Force. Ensure all records generated as a result of processes prescribed in this publication adhere to AFI 33-322, *Records Management and Information Governance Program*, and are disposed IAW the Air Force Records Disposition Schedule, which is located in the Air Force Records Information Management System. Refer recommended changes and questions about this publication to the OPR using the DAF Form 847, *Recommendation for Change of Publication*; route DAF Forms 847 from the field through the appropriate functional chain of command. This publication may be supplemented at any level, but all supplements must be routed to the OPR of this publication for coordination prior to certification and approval. The authorities to waive wing or unit level requirements in this publication are identified with a Tier ("T-0, T-1, T-2, T-3") number following the compliance statement. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval authority, or alternately, to the publication OPR for non-tiered compliance items. Do not deviate from the policies and guidance in this ACCMAN under normal circumstances, except for safety or when necessary to protect the crew or aircraft from a situation not covered by this document and immediate action is required. Report deviations or exceptions without a waiver through channels to Air Combat Command Standardization and Evaluations (ACC/A3TV). Non-tiered compliance statements in this

document are waiverable only by the Air Combat Command Director of Operations (ACC/A3). Submit waiver requests through ACC/A3TV.

SUMMARY OF CHANGES

This publication has been substantially revised and needs to be reviewed. Pilot evaluation requirements and associated table have been revised. Requirement for Mission System Operators to accomplish boldface has been deleted. Organizations have been updated to reflect current organizational structures.

Chapter 1—ROLES AND RESPONSIBILITIES **3**

1.1. Flight Examiners (FEs): 3

1.2. Examinees: 3

1.3. General Information. 3

1.4. Grading Instructions. 4

1.5. Emergency Procedures Evaluation (EPE). 4

Chapter 2—EVALUATION REQUIREMENTS **6**

2.1. General: 6

2.2. Instrument/Qualification (INSTM/QUAL) Evaluation. 6

Table 2.1. Pilot Evaluation Criteria. 7

2.3. Mission (MSN) Evaluation. 7

2.4. Instructor (INSTR) Evaluation. 7

Table 2.2. Pilot Evaluation Requirements. 8

Table 2.3. Mission System Operator Evaluation Requirements. 9

Chapter 3—EVALUATION CRITERIA **10**

3.1. General Grading Standards: 10

3.2. Pilot General: 10

3.3. Pilot Instrument: 17

3.4. Pilot Mission: 18

3.5. Mission System Operator Criteria: 19

Attachment 1—GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION **22**

Chapter 1

ROLES AND RESPONSIBILITIES

1.1. Flight Examiners (FEs):

1.1.1. Should exercise judgment when assigning subjective area grades, when assigning the overall qualification level, and when evaluating in situations not covered explicitly by this document.

1.1.2. Will brief the examinee on the purpose, conduct, and extent of each evaluation.

1.1.3. May assist in evaluation mission planning/briefing as tasked by the examinee.

1.1.4. Will apply the grading criteria contained in **Chapter 3**, as applicable.

1.1.5. Will debrief the examinee's overall rating, specific deviations, area grades assigned (if other than qualified), and any required additional training, at a minimum.

1.2. Examinees:

1.2.1. Will accomplish required flight planning IAW the crew position for the evaluation, furnishing FEs a copy of necessary mission data and mission materials, if appropriate.

1.2.2. Will brief the mission commensurate with the position in which the evaluation is being received. (Aircraft Commander, co-pilot, Instructor Pilot (IP) or FE).

1.3. General Information.

1.3.1. General. FEs will evaluate Cockpit/ Crew Resource Management (CRM) skills IAW the provisions of AFMAN 11-290, *Cockpit/Crew Resource Management and Threat & Error Management Program*; AFMAN 11-290_ACCSUP, *Cockpit/Crew Resource Management and Threat and Error Management Program*; AFMAN 11-202V2 and this volume.

1.3.2. FEs will use the evaluation criteria contained in **Chapter 3** for conducting all flight and Emergency Procedures Evaluations (EPEs). To ensure standard and objective evaluations, FEs will be thoroughly familiar with the prescribed evaluation criteria.

1.3.3. The FE may fly in any seat, including the jump seat, which will best enable the FE to conduct a thorough evaluation. The FE may fly as a primary crewmember on any evaluation.

1.3.4. The FE will brief the examinee on the purpose of the evaluation and how it is to be conducted prior to flight. The examinee will accomplish required flight planning IAW the flight position during the evaluation. Furnish Higher Headquarters FEs (and unit FEs as determined locally) a copy of necessary mission data, and mission materials.

1.3.5. Required areas are shown in **Table 2.2** and **Table 2.3**. When a required in-flight area is impossible to evaluate, use an alternate method (e.g., in a simulator/Aircrew Training Device (ATD)/Cockpit Procedures Trainer (CPT)) or by oral evaluation in order to complete the evaluation. FEs will document the reason why required area(s) were not evaluated inflight and the alternate method of evaluation used in the Comments portion of the AF Form 8, *Certificate of Aircrew Qualification*.

1.3.6. Critical Areas. Critical areas require adequate accomplishment by the aircrew member in order to successfully achieve the mission objectives. If an aircrew member receives an

unqualified grade in any of the critical areas identified by this volume, an overall unqualified grade is assigned. Unsatisfactory performance in any critical area or critical subarea results in qualification level of "Q-3." Critical areas are identified by an "*" in the "Notes" column of **Table 2.2** and **Table 2.3**.

1.3.7. The FE will thoroughly debrief all aspects of the flight. This debrief includes the examinee's overall rating, specific deviations, area grades assigned (if other than qualified) and any required additional training.

1.4. Grading Instructions. Standards and performance parameters are contained in AFMAN 11-202V2 and this volume.

1.4.1. The FE will base tolerances for inflight parameters on conditions of smooth air and a stable aircraft. Do not consider momentary deviations from tolerances, provided the examinee applies prompt corrective action and such deviations do not jeopardize flying safety. FEs will consider cumulative deviations when determining the overall grade.

1.4.2. When grading criteria specify that airspeed be evaluated and the flight manual lists only a minimum/maximum/recommended airspeed for that area, examinees will brief the desired airspeed.

1.4.3. FEs will compare examinee performance for each area accomplished during the evaluation with the standards provided in this volume and assign an appropriate grade for the area. Derive the overall flight evaluation grade from the area grades based on a composite for the observed events and tasks IAW this volume.

1.4.4. Determine the overall grade based on this grading volume and FE judgment.

1.5. Emergency Procedures Evaluation (EPE). In order of preference, conduct the EPE in a flight simulator/Unit Training Device (UTD), CPT, or verbally. Only conduct a verbal EPE if a flight simulator/UTD or CPT is not available or not configured appropriately for the evaluation. If this option is utilized, it is noted on the AF Form 8 in the additional comments section. Mission Crew, Telemetry Relay System (TMRS) and Sea Surveillance Radar (SSR), do not use UTD/CPT. Grading criteria for each required item are listed in **Chapter 3**. Evaluation areas are commensurate with examinee's training level and event qualifications and should highlight current items of interest such as recently discovered aircraft anomalies or unusual accident investigation results.

1.5.1. Include the following items on all pilot EPEs:

1.5.1.1. Aircraft general knowledge.

1.5.1.2. Emergency Procedures (EP). Evaluate a minimum of two EPs per phase of flight (e.g., pre-takeoff, takeoff, cruise, and landing). All Bold Face will be evaluated.

1.5.1.3. Crew coordination (if applicable).

1.5.1.4. Unusual attitude recoveries. This fulfills unusual attitude recoveries requirement for pilot Instrument (INSTM)/Qualification (QUAL) evaluations.

1.5.2. Include the following additional items on pilot EPEs as a requisite for the INSTM and/or QUAL evaluation:

1.5.2.1. AFMAN 11-202, Volume 3, *Flight Operations*. Evaluate use of standby/emergency instruments.

1.5.2.2. Alternate/Divert Airfields. Evaluate a minimum of one approach at other than home base.

1.5.3. Include the following items on all Mission System Operator EP evaluations:

1.5.3.1. Emergency equipment location.

1.5.3.2. EP. Evaluate EP during prior-to-engine start, engine start, taxi, takeoff, in-flight, and landing.

1.5.3.3. Crew coordination.

1.5.4. Place examinees receiving an overall unqualified grade in supervised status until recommended additional training is completed and/or a reevaluation is successfully accomplished. Accomplish additional training and reevaluations IAW AFMAN 11-202V2.

1.5.5. Normally, EPEs are conducted separately for the pilot INSTM/QUAL and pilot Mission (MSN) evaluations. In situations where INSTM/QUAL and MSN evaluation eligibility zones coincide, a single EPE may be administered to fulfill the requisites for the combined INSTM/QUAL/MSN evaluations. The combined EPE must be of sufficient scope and length to ensure all required areas for each evaluation are accomplished.

1.5.6. Use the following grading criteria to grade individual items on EPEs:

1.5.6.1. Q. Performance is correct. Quickly recognizes and corrects errors.

1.5.6.2. Q-. Performance is safe but indicates limited proficiency. Makes errors of omission or commission.

1.5.6.3. U. Performance is unsafe or indicates lack of knowledge or ability.

Chapter 2

EVALUATION REQUIREMENTS

2.1. General:

2.1.1. Follow the guidelines set in AFMAN 11-202V2 for all evaluations. Evaluation requirements are depicted in **Table 2.2** and **Table 2.3**.

2.1.2. Areas indicated with an "R" are required items for that evaluation. A required area is a specific area that must be evaluated to complete the evaluation. All required areas must be included in the flight evaluation profile. However, if impossible to accomplish a required area in-flight, the FE may elect to evaluate the area(s) by an alternate method (e.g., simulator/UTD, CPT, verbally), in order to complete the evaluation. If the FE determines the required item cannot be adequately evaluated by an alternate method, an additional flight is required to complete the evaluation.

2.2. Instrument/Qualification (INSTM/QUAL) Evaluation. A mission flown according to instrument flight rules (to the maximum extent practical) best fulfills the objective of the pilot INSTM/QUAL evaluation. To the maximum extent possible, FEs will ensure this evaluation includes approaches at airfields other than home. This evaluation may be administered on any compatible training mission with the approval of the unit commander. Mission System Operator QUAL evaluations should be conducted in conjunction with mission sorties. If the FE determines required MSN evaluation events can be accomplished on other than mission sorties (e.g., beam cut, independent SSR sortie), the FE may elect to conduct the evaluation with concurrence of the Operations Group Commander. Minimum ground phase requisites are:

2.2.1. ***[Pilot only]*** Instrument Refresher Course (IRC) Training (IAW AFMAN 11-210, *Instrument Refresher Program (IRP)*).

2.2.2. ***[Pilot only]*** INSTM examination.

2.2.3. Closed and open book qualification examinations.

2.2.4. EPE.

2.2.5. ***[Pilot only]*** Bold Face Written Exam. Answers must contain all Bold Face items in proper sequence.

2.2.6. The general criteria listed in **Table 2.1** apply during all phases of flight except as noted for specific events and instrument final approaches:

Table 2.1. Pilot Evaluation Criteria.

General		
Q	Altitude	**+/ -** 100 feet
	Airspeed	**+/ -** 10 knots (kts)
	Course	**+/ -** 5 degrees/2 Nautical Miles (NM) (whichever is greater)
	Tactical Air Navigation (TACAN) Arc	**+/ -** 2 NM
Q-	Altitude	**+/ -** 200 feet
	Airspeed	**+/ -** 15 kts
	Course	10 degrees/3 NM (whichever is greater)
	TACAN Arc	**+/ -** 3 NM
U		Exceeded Q- limits
INSTRUMENT APPROACHES		
Q	Altitude	Prior to Final Approach: +/- 100 feet Final Approach: -0/+50 feet at MDA
	Airspeed	Prior to Final Approach: +/- 10 kts Final Approach: -5/+10 kts
	Course	+/- 5 degrees or +/- 1 dot deflection
	TACAN Arc	+/- 2 NM
Q-	Altitude	Prior to Final Approach: +/- 200 feet Final Approach: -0/+50 feet at MDA
	Airspeed	Prior to Final Approach: +/- 15 kts Final Approach: -10 + 15 kts
	Course	+/- 10 degrees or +/- 1 ½ dot deflection
	TACAN Arc	+/- 3 NM
U		Exceeded Q- limits

2.3. Mission (MSN) Evaluation. Scenarios that represent unit Designated Operational Capability (DOC) tasking satisfy the requirement of this evaluation. The profiles are designed to evaluate the training/flight position/special qualifications as well as basic airmanship of the examinee. FEs must ensure initial (INIT) pilot mission evaluations are flown in the primary DOC of the unit. MSN evaluations will include examinee knowledge of operations as described per **Chapter 3** evaluation criteria. INIT QUAL pilots may receive INIT MSN evaluations after completing an INSTM/QUAL evaluation. Recurring mission evaluations will normally be combined with recurring INSTM/QUAL evaluations. The FE will only evaluate Basic Mission Capable pilots on those missions routinely performed. FEs will only evaluate examinees on those areas for which they are qualified.

2.4. Instructor (INSTR) Evaluation. INSTR evaluations are conducted IAW AFMAN 11-202V2.

Table 2.2. Pilot Evaluation Requirements.

AREA	NOTES	AREA TITLE	INSTM/ QUAL	MSN
GENERAL				
1		MISSION PLANNING	R	R
2		BRIEFING	R	R
3		PRE-TAKEOFF	R	
4		TAKEOFF	R	
5		DEPARTURE	R	
6		LEVEL OFF	R	
7		CRUISE/NAVIGATION	R	R
302		CHECKLIST PROCEDURES/USE	R	R
11		FUEL MANAGEMENT	R	R
12		COMM/TRANSPONDER	R	R
37		CREW COORDINATION/CRM	R	R
18		DESCENT	R	
19		GO-AROUND	R	
70		SIMULATED SINGLE ENGINE PATTERN/APPROACH/LANDING	R	
74		SIMULATED SINGLE ENGINE GO-AROUND	R	
301	*	EMERGENCY PROCEDURES	R	
23		VFR PATTERN	R	
25		LANDING	R	
75		TOUCH-AND-GO LANDING	R	
26		AFTER LANDING	R	
28		DEBRIEF	R	R
29		GENERAL KNOWLEDGE	R	R
30	*	AIRMANSHIP	R	R
31	*	SAFETY	R	R
32	*	AIRCREW DISCIPLINE	R	R
INSTRUCTOR ONLY				
33	1	INSTRUCTIONAL ABILITY	R	R
INSTRUMENTS				
61		HOLDING	D	
63		INSTRUMENT APPROACHES		

64		NON-PRECISION APPROACH	R	
65	2	PRECISION APPROACH	R	
67		MISSED APPROACH/CLIMB OUT	R	
68	5	CIRCLING	R	
743		RANGE PATROL		R
742	4	TM COLLECTION/RELAY		R
741	3	UHF RELAY		R

*Denotes Critical Area.

Notes:

1. Instructional ability is required if crewmember is a qualified instructor.
2. Both a PAR and an ILS should be evaluated if equipment and facilities are available. However, if facilities or equipment are not available, the flight evaluation may be completed as long as one precision approach is flown.
3. UHF relay may be verbally evaluated if not scheduled.
4. TM collection may be simulated in flight.
5. A circling approach is only required and will be accomplished in lieu of a VFR pattern when it is not available due to weather.

Table 2.3. Mission System Operator Evaluation Requirements.

AREA	NOTES	AREA TITLE	QUAL/MSN
GENERAL			
1		MISSION PLANNING	R
2		MISSION BRIEFING	R
48		MISSION EQUIPMENT PREFLIGHT	R
755		MISSION EQUIPMENT OPERATION	R
736		EQUIPMENT TROUBLESHOOTING	R
28		MISSION DEBRIEFING	R
30	*	AIRMANSHIP	R
31	*	SAFETY	R
32	*	AIRCREW DISCIPLINE	R
29		GENERAL KNOWLEDGE	R
301	*	EMERGENCY PROCEDURES	R
INSTRUCTOR ONLY			
33	1	INSTRUCTIONAL ABILITY	R

*Denotes Critical Area.

Note:

1. Instructional ability is required if crewmember is a qualified instructor.

Chapter 3

EVALUATION CRITERIA

3.1. General Grading Standards:

3.1.1. The grading criteria in this chapter are divided into three sections: General, Instrument, and Mission. Use all sections for criteria applicable to the events performed on the evaluation.

3.1.2. Where major areas are divided into subareas, only one grade is assigned to the major areas. Discrepancies on the back of the AF Form 8 are annotated by subarea.

3.2. Pilot General:

3.2.1. Area 1--Mission Planning:

3.2.1.1. Mission Planning:

3.2.1.1.1. **Q.** Developed a sound plan to accomplish the mission. Checked all factors applicable to flight (e.g., Flight Crew Information File (FCIF), weather, Notice to Air Missions (NOTAM), alternate airfields, flight logs or computer flight plans, performance data, fuel requirements, and charts). Was aware of available alternatives if flight could not be completed as planned. Read and initialed for all items in the FCIF. Was prepared at briefing time.

3.2.1.1.2. **Q-.** Minor errors or omissions that did not detract from mission effectiveness. Marginal knowledge of performance capabilities or approved operating procedures in some areas.

3.2.1.1.3. **U.** Major errors or omissions that would preclude safe/effective mission accomplishment. Faulty knowledge of operating data or procedures. Not prepared at briefing time. Did not sign off FCIF prior to flight.

3.2.1.2. Publications:

3.2.1.2.1. **Q.** Publications were current and usable for any of the unit's taskings. Contained only minor deviations, omissions and/or errors.

3.2.1.2.2. **Q-.** Publications contained deviations, omissions, and/or errors; however, contained everything necessary to effectively accomplish the mission and did not compromise safety of flight.

3.2.1.2.3. **U.** Not up to "Q-" standards. Contained major deviations, omissions, and/or errors.

3.2.2. Area 2--Briefing:

3.2.2.1. Organization:

3.2.2.1.1. **Q.** Well organized and presented in a logical sequence. Concluded briefing in time to allow for preflight of personal equipment and aircraft.

3.2.2.1.2. **Q-.** Events out of sequence, hard to follow, some redundancy.

3.2.2.1.3. **U.** Confusing presentation. Did not allow time for preflight of personal equipment and aircraft.

3.2.2.2. Presentation:

3.2.2.2.1. **Q.** Presented briefing in a professional manner. Crewmembers clearly understood mission requirements.

3.2.2.2.2. **Q-.** Dwelled on non-essential mission items.

3.2.2.2.3. **U.** Redundant throughout briefing. Lost interest of crewmembers. Presentation created doubts or confusion.

3.2.2.3. Mission Coverage:

3.2.2.3.1. **Q.** Established objectives for the mission. Presented all training events and effective techniques discussed for accomplishing the mission.

3.2.2.3.2. **Q-.** Omitted some minor training events. Limited discussion of techniques.

3.2.2.3.3. **U.** Did not establish objectives for the mission. Omitted major training events or did not discuss techniques.

3.2.2.4. Crewmember Consideration:

3.2.2.4.1. **Q.** Considered the abilities of all crewmembers. Briefed corrective action from previous mission and probable problem areas when appropriate.

3.2.2.4.2. **Q-.** Did not consider all crewmembers' abilities. Did not identify probable problem areas.

3.2.2.4.3. **U.** Ignored crewmembers' abilities and past problem areas.

3.2.3. Area 3--Pre-Takeoff:

3.2.3.1. **Q.** Established and adhered to station, start engine, taxi, and takeoff times to assure a thorough preflight, check of personnel equipment, crew/ passenger briefing, etc. Performed all checks and procedures prior to takeoff IAW approved checklists and applicable directives.

3.2.3.2. **Q-.** Minor procedural deviations which did not detract from mission effectiveness.

3.2.3.3. **U.** Omitted major item(s) of the appropriate checklist. Major deviations in procedures which would preclude safe mission accomplishment. Failed to accurately determine readiness of aircraft for flight. Crew errors directly contributed to a late takeoff which degraded the mission or made it non-effective.

3.2.4. Area 4--Takeoff:

3.2.4.1. **Q.** Smooth aircraft control throughout takeoff. Performed takeoff IAW flight manual procedures and techniques.

3.2.4.2. **Q-.** Minor flight manual procedural or technique deviations. Some under or over-control at lift-off.

3.2.4.3. **U.** Lift-off potentially dangerous. Exceeded aircraft/systems limitations. Failed to establish proper climb attitude. Over-controlled aircraft resulting in excessive deviations from intended flight path.

3.2.5. Area 5--Departure:

3.2.5.1. **Q.** Performed departure as published/directed and complied with all restrictions.

3.2.5.2. **Q-.** Minor deviations in airspeed and navigation occurred during completion of departure.

3.2.5.3. **U.** Failed to comply with published/directed departure instructions. Accepted clearance which could not be complied with. Exceeded limits of Q-.

3.2.6. Area 6--Level Off:

3.2.6.1. **Q.** Leveled off smoothly. Promptly established proper cruise airspeed.

3.2.6.2. **Q-.** Level-off was erratic. Slow in establishing proper cruise airspeed.

3.2.6.3. U. Level-off was erratic. Exceeded Q- limits. Excessive delay or failed to establish proper cruise airspeed. Failed to reset altimeter, as required.

3.2.7. Area 7--Cruise/Navigation:

3.2.7.1. **Q.** Demonstrated satisfactory capability to navigate using all available means. Used appropriate navigation procedures. Ensured navaids were properly tuned, identified, and monitored. Complied with clearance instructions. Aware of position at all times. Remained within the confines of assigned airspace.

3.2.7.2. **Q-.** Minor errors in procedures/use of navigation equipment. Some deviations in tuning, identifying, and monitoring navaids. Slow to comply with clearance instructions. Had some difficulty establishing exact position and course.

3.2.7.3. **U.** Made major errors in procedures or use of navigation equipment to the extent that position was unreliable. Exceeded limits of Q-.

3.2.8. Area 302--Checklist Procedures/Use:

3.2.8.1. **Q.** Performed all checks as required.

3.2.8.2. **Q-.** Minor deviations or omissions during checks. Did not detract from mission accomplishment. Tended to make minor checklist deviations including incorrect or untimely responses or used the incorrect checklist.

3.2.8.3. **U.** Did not perform checks or monitor systems to the degree that an emergency condition would have developed if allowed to continue uncorrected. Consistently omitted checklist items or used the wrong checklist. Unable to identify the correct checklist to use for a given situation.

3.2.9. Area 11--Fuel Management:

3.2.9.1. **Q.** Actively monitored fuel throughout the mission. Complied with all established fuel requirements.

3.2.9.2. **Q-.** Errors in fuel management procedures which did not preclude mission accomplishment.

3.2.9.3. **U.** Failed to monitor fuel status or comply with established fuel requirements. Poor fuel management precluded mission accomplishment.

3.2.10. Area 12--Communication (COMM)/Transponder:

3.2.10.1. **Q.** Complete knowledge and compliance with COMMs procedures. Transmissions concise with proper terminology.

3.2.10.2. **Q-.** Occasional deviations from correct procedures that required retransmissions or resetting codes. Slow in initiating required actions. Transmissions contained extraneous matter, were not in proper sequence, or used nonstandard terminology.

3.2.10.3. **U.** Incorrect procedures or poor performance precluded mission accomplishment or jeopardized safety.

3.2.11. Area 37—Crew Coordination/Crew Resource Management (CRM). See AFMAN 11-290; and use AF Form 4031, *CRM/TEM Skills Criteria Training/Evaluation*, as a reference.

3.2.11.1. **Q.** Effectively coordinated with other crewmember without misunderstanding. Effective use of CRM.

3.2.11.2. **Q-.** Coordinated with other crewmember with minor misunderstanding. Crew communications were not clear or concise. CRM was the minimum acceptable.

3.2.11.3. **U.** Breakdown in coordination with other crewmember precluded mission accomplishment or jeopardized safety. CRM was lacking and the mission accomplishment was severely degraded.

3.2.12. Area 18--Descent:

3.2.12.1. **Q.** Performed descent as directed. Complied with all restrictions.

3.2.12.2. **Q-.** Performed descent as directed with minor deviations.

3.2.12.3. **U.** Failed to comply with published/directed descent instructions. Exceeded Q-standards.

3.2.13. Area 19--Go-Around:

3.2.13.1. **Q.** Initiated and performed go-around promptly IAW published procedures and directives. No course/heading deviation.

3.2.13.2. **Q-.** Slow to initiate go-around or procedural steps. Control inputs were in correct direction but some over/under control.

3.2.13.3. **U.** Did not self-initiate go-around when appropriate or directed. Techniques unsafe or applied incorrect procedures. Exceeded Q- parameters.

3.2.14. Area 70--Simulated Single Engine Pattern/Approach/Landing:

3.2.14.1. **Q.** Complied with all flight manual and operational procedures. Approach and landing were well controlled, on centerline, on glide-path and in the touchdown zone.

3.2.14.2. **Q-.** Minor procedural errors. Minor aircraft airspeed control. Errors did not detract from safe handling of the situation. Approach and landing were safe, but with minor control problems or deviations. Landing was in the touchdown zone.

3.2.14.3. **U.** Did not comply with applicable procedures. Erratic control compounded problems associated with the emergency. Flew an approach which was incompatible with the simulated emergency. Did not adjust approach for simulated emergency. Landing was unsafe.

3.2.15. Area 74--Simulated Single Engine Go-Around:

3.2.15.1. **Q.** Initiated and performed go-around promptly IAW published procedures and directives. No course/heading deviation.

3.2.15.2. **Q-.** Slow to initiate go-around or procedural steps. Control inputs were in correct direction but some over/under control.

3.2.15.3. **U.** Did not self-initiate go-around when appropriate or directed. Techniques unsafe or applied incorrect procedures. Exceeded Q- parameters.

3.2.16. Area 301--Emergency Procedures (EP):

3.2.16.1. **Q.** Performed all appropriate critical action and emergency procedure checklist items. Correct response to Bold Face (if applicable) and non-Bold Face emergency situations. Effectively used checklist. Coordinated proper crew actions.

3.2.16.2. **U.** Omitted or performed critical action items in improper sequence. Incorrect EP applied to scenario. Incorrect response for Bold Face (if applicable) emergency. Unable to analyze problems or take corrective action. Did not use checklist or lacks acceptable familiarity with its arrangement or contents.

3.2.17. Area 23—Visual Flight Rules (VFR) Pattern:

3.2.17.1. **Q.** Performed traffic patterns/approaches IAW procedures and techniques outlined in the flight manual, operational procedures and local directives. Aircraft control was positive and smooth.

3.2.17.2. **Q-.** Performed patterns/approaches with minor deviations to procedures and techniques outlined in the flight manual, operational procedures and local directives. Aircraft control was not consistently smooth, but safe. Variations in runway alignment. Slow to correct to proper/briefed airspeed.

3.2.17.3. **U.** Approaches not performed IAW procedures and techniques outlined in the flight manual, operational procedures, and local directives. Erratic aircraft control. Large deviations in runway alignment. Exceeded Q- Parameters.

3.2.18. Area 25--Landing:

3.2.18.1. **Q.** Landing was on centerline in the touchdown zone.

3.2.18.2. **Q-.** Minor control problems but was safe and in the touchdown zone.

3.2.18.3. **U.** Landing was unsafe or outside the touchdown zone.

3.2.19. Area 75--Touch-and-Go Landing:

3.2.19.1. **Q.** Maintained smooth, positive aircraft control throughout landing, landing roll and takeoff. Performed the maneuver as published/directed.

3.2.19.2. **Q-.** Made minor deviations from published procedures without affecting safety of flight. Control was slightly rough or erratic. Was hesitant in application of procedures/corrections.

3.2.19.3. **U.** Maneuver was potentially dangerous. Failed to establish proper aircraft configuration. Over controlled the aircraft resulting in excessive deviation from runway centerline.

3.2.20. Area 26--After Landing:

3.2.20.1. **Q.** Appropriate after landing, taxi and engine shutdown procedures IAW flight manual and applicable directives. Completed all forms accurately.

3.2.20.2. **Q-.** Minor deviations or omissions noted in performance of after landing, taxi or engine shutdown procedures which did not jeopardize safety. Required forms completed with minor errors.

3.2.20.3. **U.** Major deviations or omissions were made in performance of the after landing, taxi or engine shutdown procedures which could have jeopardized safety. Required forms not completed.

3.2.21. Area 28--Debrief:

3.2.21.1. **Q.** Thoroughly debriefed the mission (or applicable portions). Compared mission results with initial objectives that were established for the mission. Debriefed deviations.

3.2.21.2. **Q-.** Limited debriefing. Did not thoroughly discuss performance in relationship to mission objectives. Did not debrief all deviations.

3.2.21.3. **U.** Did not debrief the mission deviations.

3.2.22. Area 29--General Knowledge:

3.2.22.1. Aircraft General:

3.2.22.1.1. **Q.** Thorough knowledge of aircraft systems, limitations, performance characteristics.

3.2.22.1.2. **Q-.** Limited knowledge of aircraft systems, limitations, performance characteristics.

3.2.22.1.3. **U.** Unsatisfactory knowledge of aircraft systems, limitations, performance characteristics.

3.2.22.2. Flight Rules/Procedures:

3.2.22.2.1. **Q.** Thorough knowledge of flight rules and procedures.

3.2.22.2.2. **Q-.** Deficiencies in-depth of knowledge.

3.2.22.2.3. **U.** Inadequate knowledge of flight rules and procedures.

3.2.23. Area 30--Airmanship:

3.2.23.1. **Q.** Executed the assigned mission in a timely, efficient manner. Included crew inputs in decision making process.

3.2.23.2. **U.** Decisions or lack thereof resulted in failure to accomplish the assigned mission. Demonstrated poor judgment to the extent that safety could have been compromised. Ignored crew inputs in decision making process.

3.2.24. Area 31--Safety:

3.2.24.1. **Q.** Aware of and complied with all safety factors required for safe aircraft operation and mission accomplishment.

3.2.24.2. **U.** Was not aware or did not comply with all safety factors required for safe operation or mission accomplishment. Did not adequately clear. Operated the aircraft in a dangerous manner.

3.2.25. Area 32--Aircrew Discipline:

3.2.25.1. **Q.** Demonstrated strict professional flight and crew discipline throughout all phases of the mission.

3.2.25.2. **U.** Failed to exhibit strict flight or crew discipline. Violated or ignored rules or regulations.

3.2.26. Area 33--Instructional Ability:

3.2.26.1. Briefing/Debriefing:

3.2.26.1.1. **Q.** Thoroughly reviewed student training folder and presented a comprehensive, instructional briefing/debriefing which encompassed all mission events. Excellent analysis of all events/maneuvers. Clearly defined objectives.

3.2.26.1.2. **Q-.** Minor errors or omissions in student training folder review, briefing/debriefing or mission critique. Occasionally unclear in analysis of events or maneuvers.

3.2.26.1.3. **U.** Major errors or omissions in student training folder review, briefing/debriefing. Analysis of events or maneuvers was incomplete, inaccurate, or confusing. Did not use reference material effectively. Briefing/debriefing below the caliber of that expected of instructors. Failed to define mission, training objectives or brief/debrief student.

3.2.26.2. Demonstration of Maneuvers:

3.2.26.2.1. **Q.** Performed required maneuvers within prescribed parameters. Provided concise, meaningful in-flight instruction. Demonstrated excellent instructor proficiency.

3.2.26.2.2. **Q-.** Performed required maneuvers with minor deviations from prescribed parameters. In-flight instruction was unclear. Weak instructor proficiency.

3.2.26.2.3. **U.** Was unable to properly perform required maneuvers. Made major procedural errors. Did not provide in-flight instruction. Did not demonstrate instructor proficiency.

3.2.26.3. Instructor Knowledge:

3.2.26.3.1. **Q.** Demonstrated in-depth knowledge of procedures, requirements, aircraft systems, performance characteristics and mission.

3.2.26.3.2. **Q-.** Deficiencies in depth of knowledge, comprehension of procedures, requirements, aircraft systems, performance characteristics and mission.

3.2.26.3.3. **U.** Unfamiliar with procedures, requirements, aircraft systems, performance characteristics, and mission. Lack of knowledge in certain areas seriously detracted from instructor effectiveness.

3.2.26.4. Training/Evaluation Forms Preparation:

3.2.26.4.1. **Q.** Completed appropriate training/evaluation records accurately. Adequately assessed and recorded performance. Comments were clear and pertinent.

3.2.26.4.2. **Q-.** Minor errors or omissions in training/ evaluation records. Comments were incomplete or slightly unclear.

3.2.26.4.3. **U.** Did not complete required forms or records. Comments were invalid, unclear or did not accurately document performance.

3.2.26.5. Ability to Instruct:

3.2.26.5.1. **Q.** Demonstrated excellent instructor/evaluator ability. Clearly defined all mission requirements and any required additional training/corrective action. Instruction/evaluation was accurate, effective, and timely. Was completely aware of aircraft/mission situation at all times.

3.2.26.5.2. **Q-.** Problems in communication or analysis degraded effectiveness of instruction/ evaluation.

3.2.26.5.3. **U.** Demonstrated inadequate ability to instruct/ evaluate. Unable to perform, teach, or assess techniques, procedures, or systems use. Did not remain aware of aircraft/mission situation at all times.

3.3. Pilot Instrument:

3.3.1. Area 61--Holding:

3.3.1.1. **Q.** Performed entry and holding procedures as published/directed.

3.3.1.2. **Q-.** Performed entry and holding procedures with minor deviations to directives that did not jeopardize safety.

3.3.1.3. **U.** Exceeded holding airspace limits or the criteria limits for Q-.

3.3.2. Area 63--Instrument Approaches:

3.3.2.1. **Q.** Performed procedures as published or directed and IAW flight manual. Smooth and timely response to controller's instructions.

3.3.2.2. **Q-.** Performed procedures with minor deviations. Slow to respond to controller's instructions.

3.3.2.3. **U.** Performed procedures with major deviations/ erratic corrections. Failed to comply with controller's instructions.

3.3.3. Area 64--Non-Precision Approach:

3.3.3.1. **Q.** Performed procedures as published/directed. Made smooth and timely corrections. Arrived at Minimum Descent Altitude (MDA) prior to or at visual descent point. Position would have permitted safe landing. Correctly computed/adjusted timing when required.

3.3.3.2. **Q-.** Performed procedure with minor deviations. Slow to make corrections. Arrived at MDA prior to or at missed approach point. Position would have permitted safe landing. Made minor errors in computing/adjusting timing when required.

3.3.3.3. **U.** Performed procedures with major deviations. Made erratic corrections. Did not arrive at MDA prior to or at missed approach point/visual descent point. Position would not have permitted safe landing. Failed to compute or adjust timing when required.

3.3.4. Area 65--Precision Approach:

3.3.4.1. **Q.** Performed procedures as published/directed. Made smooth and timely corrections. Position would have permitted a safe landing.

3.3.4.2. **Q-.** Performed procedures with minor deviations. Was slow to make proper corrections. Position would have permitted a safe landing.

3.3.4.3. **U.** Performed procedures with major deviations. Made erratic corrections. Did not respond to controller instructions. Position would not have permitted a safe landing. Exceeded limits of Q-.

3.3.5. Area 67--Missed Approach/Climb Out:

3.3.5.1. **Q.** Executed missed approach as published/ directed.

3.3.5.2. **Q-.** Executed missed approach with minor deviations. Was slow to comply with published procedures, controller's instructions or flight manual procedures.

3.3.5.3. **U.** Executed missed approach with major deviations. Failed to comply with published procedures, controller's instructions or flight manual procedures.

3.3.6. Area 68--Circling:

3.3.6.1. **Q.** Complied with all procedures/controller instructions. Remained within the specified distance. Achieved proper displacement. Completed circle to final without over/undershoot.

3.3.6.2. **Q-.** Complied with all procedures/controller instructions but aircraft control was not smooth or applied improper techniques for achieving proper displacement. Remained within specified distance but circle resulted in over/undershoot.

3.3.6.3. **U.** Failed to follow procedures or controller instructions. Did not remain within the specified distance. Lost sight of the field but did not execute a missed approach. Did not meet the parameters of Q-.

3.4. Pilot Mission:

3.4.1. Area 743--Range Patrol:

3.4.1.1. **Q.** Performed range patrol IAW local procedures. Range Patrol was accomplished efficiently.

3.4.1.2. **Q-.** Performed range patrol IAW procedures. Range patrol was not accomplished efficiently but was still accomplished.

3.4.1.3. **U.** Range patrol was not IAW procedures.

3.4.2. Area 742--Telemetry Collection/Relaying: (Assumes no hardware malfunctions limiting telemetry collection/relay.)

3.4.2.1. **Q.** Flew aircraft in best position to collect telemetry. Coordinated with the telemetry operator.

3.4.2.2. **Q-.** Telemetry was collected/relayed to satisfy the mission, but telemetry collection was sacrificed due to aircraft position or poor coordination with telemetry operator.

3.4.2.3. **U.** Telemetry was not collected/relayed to satisfy the mission due to the way the aircraft was positioned or lack of coordination with the telemetry operator. Did not meet Q- criteria.

3.4.3. Area 741—Ultra High Frequency (UHF) Relay:

3.4.3.1. **Q.** The aircraft was in the best position for UHF relay. Coordination with the system operator was sufficient to satisfy mission.

3.4.3.2. **Q-.** The aircraft was in position for UHF relay. Lack of coordination with the system operator resulted in deviations but capable of satisfying mission requirements.

3.4.3.3. **U.** The aircraft was in the wrong location or coordination with the system operator did not allow support of the UHF relay and did not satisfy the mission.

3.5. Mission System Operator Criteria:

3.5.1. Area 1--Mission Planning:

3.5.1.1. **Q.** Developed a sound plan to accomplish the mission. Checked all factors applicable to flight IAW applicable directives. Aware of alternatives available if flight cannot be completed as planned. Prepared at briefing time.

3.5.1.2. **Q-.** Same as above, except minor error(s) or omission(s) that did not detract from mission effectiveness. Demonstrated limited knowledge of performance capabilities or approved operating procedures/rules in some areas.

3.5.1.3. **U.** Made major error(s) or omission(s) that would have prevented a safe or effective mission. Displayed faulty knowledge of operating data or procedures. Not prepared at briefing time.

3.5.2. Area 2--Mission Briefing:

3.5.2.1. **Q.** Briefing was logical and professionally delivered. Mission elements clearly understood by all flight members. Mission suffered no degradation due to mission briefing.

3.5.2.2. **Q-.** Briefing was clear but disjointed and dwelled on non-mission essential elements. Mission suffered minor deviations due to briefing, but overall mission accomplishment was possible.

3.5.2.3. **U.** Briefing unclear and led to confusion of flight members regarding mission elements. Mission suffered unrecoverable degradations due to mission briefing.

3.5.3. Area 48--Mission Equipment Preflight:

3.5.3.1. **Q.** Performed all required TMRS or SSR pre-flight checks efficiently and without hesitation.

3.5.3.2. **Q-.** Pre-flight of equipment was essentially correct but with minor deviations that detracted from the mission.

3.5.3.3. **U.** Performed incorrect or inadequate preflight of TMRS or SSR that resulted in severe mission degradation. Did not perform a TMRS or SSR preflight.

3.5.4. Area 755--Mission Equipment Operation:

3.5.4.1. **Q.** Telemetry (TM) Examinee collected/relayed data without operator-induced interruption, SSR examinee collaborated with the Range Safety Officer (RSO) establishing a shoot container per mission requirements.

3.5.4.2. **Q-.** TM examinee collected/relayed data accomplishing mission requirement but was delayed due to self-inflicted errors. SSR examinee delayed the RSO's construction of the shoot container due to lack of awareness or slow system configuration but did not hinder mission success.

3.5.4.3. **U.** TM examinee did not detect improper system configuration, thereby failing to collect/relay data. SSR examinee prevented or considerably delayed the RSO from constructing a shoot container, inhibiting mission success.

3.5.5. Area 736--Equipment Troubleshooting:

3.5.5.1. **Q.** Examinee analyzed TMRS or SSR system malfunctions and applied corrective action without hesitation.

3.5.5.2. **Q-.** Limited debriefing. Corrective actions to TMRS or SSR system malfunctions were delayed but did not hinder mission success.

3.5.5.3. **U.** TM examinee did not respond to or apply corrective actions to system malfunction, inhibiting collection/relay of data. SSR examinee did not respond to nor correct system malfunction impeding shoot track construction, resulting in E-9 SSR mission non-effective.

3.5.6. Area 28--Mission Debriefing:

3.5.6.1. **Q.** Thoroughly debriefed applicable portions of the mission and addressed deviations. Offered corrective guidance as appropriate.

3.5.6.2. **Q-.** Limited debriefing. Did not thoroughly discuss performance in relationship to mission objectives. Did not debrief all deviations.

3.5.6.3. **U.** Did not debrief mission deviations or offer corrective guidance.

3.5.7. Area 30--Airmanship:

3.5.7.1. **Q.** Executed the assigned mission in a timely, efficient manner. Included crew inputs in decision making process.

3.5.7.2. **U.** Decisions or lack thereof resulted in failure to accomplish the assigned mission. Demonstrated poor judgment to the extent that safety could have been compromised. Ignored crew inputs in decision making process.

3.5.8. Area 31--Safety:

3.5.8.1. **Q.** Aware of and complied with all safety factors required for safe aircraft operation and mission accomplishment.

3.5.8.2. **U.** Was not aware or did not comply with all safety factors required for safe operation or mission accomplishment. Did not adequately clear. Operated the aircraft in a dangerous manner.

3.5.9. Area 32--Aircrew Discipline:

3.5.9.1. **Q.** Demonstrated strict professional flight and crew discipline throughout all phases of the mission.

3.5.9.2. **U.** Failed to exhibit strict flight or crew discipline. Violated or ignored rules or regulations.

3.5.10. Area 29--General Knowledge:

3.5.10.1. **Q.** Thorough knowledge of mission systems capabilities and limitations.

3.5.10.2. **Q-.** Limited knowledge of mission systems capabilities and limitations.

3.5.10.3. **U.** Unsatisfactory knowledge of mission systems capabilities and limitations.

3.5.11. Area 33 ***[MSO only]***--Instructional Ability:

3.5.11.1. **Q.** Demonstrated excellent instructor/evaluator ability. Clearly defined all mission requirements and any required additional training/corrective action. Instruction/evaluation was accurate, effective, and timely. Was completely aware of aircraft/mission situation at all times.

3.5.11.2. **Q-.** Demonstrated marginal ability to instruct/evaluate. Provided unclear guidance to student. Instruction/evaluation was not at times inaccurate, ineffective, and/or not timely. Fell behind aircraft, student, and/or instruction.

3.5.11.3. **U.** Demonstrated inadequate ability to instruct/evaluate. Unable to perform, teach, or assess techniques, procedures, or systems use. Did not remain aware of aircraft/mission situation at all times.

DAVID G. SHOEMAKER, Brig Gen, USAF
Director of Operations

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 33-322, *Records Management and Information Governance Program*, 23 March 2020

AFMAN 11-202V2, *Aircrew Standardization and Evaluation Program*, 30 August 2021

AFMAN 11-202V3, *Flight Operations*, 10 January 2022

AFMAN 11-210, *Instrument Refresher Program (IRP)*, 21 December 2021

AFMAN 11-290, *Cockpit/Crew Resource Management and Threat & Error Management Program*, 25 October 2021

AFMAN 11-290_ACCSUP, *Cockpit/Crew Resource Management and Threat & Error Management Program*, 10 June 2022

Prescribed Forms

None

Adopted Forms

AF Form 8, *Certificate of Aircrew Qualification*

AF Form 4031, *CRM/TEM Skills Criteria Training/Evaluation Form*

DAF Form 847, *Recommendation for Change of Publication*

Abbreviations and Acronyms

ACC—Air Combat Command

ATD—Aircrew Training Device

COMM—Communication

CPT—Cockpit Procedures Trainer

CRM—Cockpit/Crew Resource Management

DOC—Designated Operational Capability

EP—Emergency Procedures

EPE—Emergency Procedures Evaluation

FE—Flight Examiner

FCIF—Flight Crew Information File

ILS—Instrument Landing System

INIT—Initial (in reference to evaluations)

INSTM—Instrument Evaluation

INSTR—Instructor Evaluation

IP—Instructor Pilot

IRC—Instrument Refresher Course

IRP—Instrument Refresher Program

kts—Knots

MDA—Minimum Descent Altitude

MSN—Mission Evaluation

NM—Nautical Miles

NOTAM—Notice to Air Mission

PAR—Precision Approach Radar

QUAL—Qualification Evaluation

SSR—Sea Surveillance Radar

RSO—Range Safety Officer

TACAN—Tactical Air Navigation

TM—Telemetry

TMRS—Telemetry Relay System

UHF—Ultra High Frequency

UTD—Unit Training Device

VFR—Visual Flight Rules

Office Symbols

ACC/A3—Air Combat Command Director of Operations

ACC/A3T—Air Combat Command Flight Operations Division

ACC/A3TV—Air Combat Command Standardization and Evaluations Branch

Terms

Deviation—Performing an action not in sequence with current procedures, directives, or regulations. Performing action(s) out of sequence due to unusual or extenuating circumstances is not considered a deviation. In some cases, momentary deviations may be acceptable; however, cumulative momentary deviations will be considered in determining the overall qualification level.

Error—Departure from standard procedures. Performing incorrect actions or recording incorrect information.

Major—Detracted from mission accomplishment.Adversely affected use of equipment or violated safety.

Minor—Did not detract from mission accomplishment.

Omission—To leave out a required action or annotation.